

CURRICULUM REVISION 2021 - PROGRAMME SCHEME													
STATE BOARD OF TECHNICAL EDUCATION, KERALA													
TEACHING AND EXAMINATION SCHEME FOR DIPLOMA PROGRAMME													
PROGRAM NAME : Mechanical Engineering											PROGRAM CODE : ME		
PATTERN: FULL TIME													
DURATION OF PROGRAMME: 3 YEARS (6 SEMESTERS)													
SEMESTER: First Semester											SCHEME: Revision 2021		
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	1001	Communication Skills in English	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	1002	Mathematics I	5	0	0	5	50	75	125	T	B	5
3.	Basic Science	1003	Applied Physics I	3	0	0	3	50	75	125	T	B	3
4.	Basic Science	1004	Applied Chemistry	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	1005	Engineering Graphics	0	0	3	3	50	75	125	D	B	1.5
6.	Basic Science	2006	Applied Physics Lab	0	0	2	2	^	^	^	P	I	^
7.	Basic Science	1007	Applied Chemistry Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	1008	Introduction to IT systems Lab	1	0	2	3	75	50	125	P	I	2
9.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	^	^	^	P	I	^
10.	Humanities & Social Sciences	1009	Sports and Yoga	0	0	2	2	75	50	125	P	I	1
				16	0	14	30						20.5

PROGRAM NAME : Mechanical Engineering										PROGRAM CODE : ME			
SEMESTER: Second Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Credits	
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		Evaluation Mode
				L	T	P	Total						
1.	Basic Science	2002	Mathematics II	4	0	0	4	50	75	125	T	B	4
2.	Basic Science	2003	Applied Physics II	3	0	0	3	50	75	125	T	B	3
3.	Common Courses	2001	Environmental Science	3	0	0	3	50	75	125	T	I	0
4.	Engineering Science	2021	Engineering Mechanics	3	0	0	3	50	75	125	T	B	3
5.	Engineering Science	2022	Manufacturing Technology	3	0	0	3	50	75	125	T	B	3
6.	Humanities & Social Sciences	2008	Communication Skills in English Lab	0	0	3	3	75	50	125	P	I	1.5
7.	Basic Science	2006	Applied Physics Lab	0	0	2	2	75	50	125	P	I	1
8.	Engineering Science	2029	Engineering Mechanics Lab	0	0	3	3	75	50	125	P	I	0
9.	Engineering Science	2028	Basic CAD Lab	0	0	3	3	75	50	125	P	I	0
10.	Engineering Science	2009	Engineering Workshop Practice	0	0	3	3	75	50	125	P	I	1.5
				16	0	14	30						17
Summer Internship I to be conducted after Semester 2													
1.	Common Courses	3009	Internship I	3 Weeks				0	75	75	P	I	2

PROGRAM NAME : Mechanical Engineering										PROGRAM CODE : ME			
SEMESTER: Third Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Programme core course	3021	Strength of Materials	2	1	0	3	50	75	125	T	B	3
2.	Programme core course	3022	Material Science and Metrology	2	1	0	3	50	75	125	T	B	3
3.	Programme core course	3023	Machine Tools	3	1	0	4	50	75	125	T	B	4
4.	Programme core course	3024	Fundamentals of Electrical Engineering	3	1	0	4	50	75	125	T	B	4
5.	Programme core course	3025	Machine Drawing	0	0	3	3	50	75	125	D	B	1.5
6.	Programme core course	3026	Material Testing and Metallography Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	3027	Mechanical Workshop III	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	3028	Electrical and Electronics Lab	0	0	3	3	75	50	125	P	B	1.5
9.	Programme core course	3029	Advanced CADD Lab	0	0	4	4	75	50	125	P	I	0
				10	4	16	30						20

PROGRAM NAME : Mechanical Engineering										PROGRAM CODE : ME			
SEMESTER: Fourth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Programme core course	4021	Thermal Engineering	3	1	0	4	50	75	125	T	B	4
2.	Programme core course	4022	Fluid Mechanics and Hydraulic Machines	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	4023	Automobile Engineering	3	1	0	4	50	75	125	T	B	4
4.	Common Courses	4001	Community Skills in Indian knowledge system	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	4024	Industrial Engineering	3	0	0	3	50	75	125	T	I	0
6.	Programme core course	4027	Thermal Engineering Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Programme core course	4028	Fluid Mechanics Lab (Mechanical)	0	0	3	3	75	50	125	P	B	1.5
8.	Programme core course	4029	Mechanical Workshop IV	0	0	3	3	75	50	125	P	B	1.5
9.	Minor project	4009	Minor Project	0	0	4	4	0	75	75	P	I	2
				14	3	13	30						18.5
Summer Internship II to be conducted after Semester 4													
1.	Common Courses	5009	Internship II	4 Weeks				0	75	75	P	I	3

PROGRAM NAME : Mechanical Engineering										PROGRAM CODE : ME			
SEMESTER: Fifth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Common Courses	5001	Industrial Management and Safety	3	1	0	4	50	75	125	T	I	0
2.	Programme core course	5021	Design Of Machine Elements	3	1	0	4	50	75	125	T	B	4
3.	Programme core course	5022	Refrigeration and Air conditioning	3	1	0	4	50	75	125	T	B	4
4.	Program Elective course	5023A	Modern Production Process	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5023B	Material Handling	4	0	0	4	50	75	125	T	B	4
	Program Elective course	5023C	Power Plant Engineering	4	0	0	4	50	75	125	T	B	4
5.	Programme core course	5027	Machine Shop Practice	0	0	3	3	75	50	125	P	B	1.5
6.	Programme core course	5028	Hydraulic Machines Lab	0	0	3	3	75	50	125	P	B	1.5
7.	Program Elective course	5029	Automotive Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Seminar	5008	Seminar	0	0	2	2	0	75	75	P	I	1
9.	Major Project	6009	Major Project	0	0	3	3	0	0	0	P	I	0
				13	3	14	30						17.5

The Student has two options to complete the sixth semester and he/she shall select any one: 1. NORMAL PATHWAY 2. INTERNSHIP PATHWAY

1. NORMAL PATHWAY

PROGRAM NAME : Mechanical Engineering

PROGRAM CODE : ME

SEMESTER: Sixth Semester

SCHEME: Revision
2021

Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Program Elective course	6021A	Mechatronics	3	1	0	4	50	75	125	T	B	4
	Program Elective course	6021B	Computer Integrated Manufacturing	3	1	0	4	50	75	125	T	B	4
	Program Elective course	6031D	Electric Vehicles	3	1	0	4	50	75	125	T	B	4
	Program Elective course	6051A	Electric and Hybrid Vehicles	4	0	0	4	50	75	125	T	B	4
3.	Open Elective course	6022A	Computer Aided Design and Manufacturing	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6022B	Operation Research	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6022C	Renewable Energy Technologies	4	0	0	4	50	75	125	T	I	4
	Open Elective course	6022D	Product Design	4	0	0	4	50	75	125	T	I	4
4.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
5.	Programme core course	6027	Production Software Lab	0	1	3	4	75	50	125	P	I	2.5
6.	Programme core course	6028	Refrigeration and Airconditioning Lab	0	0	3	3	75	50	125	P	I	1.5
7.	Program Elective course	6029	Advanced Machine Tools Lab	0	0	3	3	75	50	125	P	B	1.5
8.	Major Project	6009	Major Project	0	0	6	6	75	50	125	P	I	4
				12	3	15	30						21.5
													120

2. INTERNSHIP PATHWAY													
PROGRAM NAME : Mechanical Engineering										PROGRAM CODE : ME			
SEMESTER: Sixth Semester										SCHEME: Revision 2021			
Sl. No.	Course Category	Course Code	Course Name	Teaching Scheme				Evaluation Scheme				Evaluation Mode	Credits
				Contact Hours/ week				CA	ESA	Total Marks	Exam Type		
				L	T	P	Total						
1.	Humanities & Social Sciences	6001	Entrepreneurship and Startup	3	1	0	4	50	75	125	T	I	4
2.	Common Courses	6002	Indian Constitution	2	0	0	2	50	75	125	T	I	0
3.	Programme core course	6027	Production Software Lab	0	1	3	4	75	50	125	P	I	2.5
4.	Programme core course	6028	Refrigeration and Airconditioning Lab	0	0	3	3	75	50	125	P	I	1.5
5.	Internship	6007	Internship	4 - 6 months per semester				250	250	500	P	I	13.5
				5	2	6	13						21.5
Teaching Scheme : L-Lecture, T-Tutorial, P- Practical Examination Type : T-Theory, P-Practical, D-Drawing Examination Mode : B-Board Level, I-Institution Level ^ Carried over to next semester													